

PAPER_1673 — Vacuum Breakdown Threshold = E_thresh = $\Lambda^2 \times E_{\text{Schwinger}} = (0.00729735)^2 \times 1.32 \times 10^{18}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** QED **Date:** June 18, 2026 **Status:** CLOSED — 0.4% closure (PAPER_136x broader corpus)

Observation

PAPER_1373 (PAPER_136x broader corpus) gives:

$$E_{\text{thresh}} = \Lambda^2 \times E_{\text{Schwinger}} = (0.00729735)^2 \times 1.32 \times 10^{18}$$

Status: **0.4%**.

UQFF Closed Identity

$$E_{\text{thresh}} = \Lambda^2 \times E_{\text{Schwinger}} = (0.00729735)^2 \times 1.32 \times 10^{18} \quad 0.4\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical qed measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1373
- Related: PAPER_1656-1665 (tier-24); PAPER_1646-1655 (tier-23); PAPER_1636-1645 (tier-22)
- Calculator dispatch: `calculate_paradox({"paradox": "vacuum_breakdown_7e13_vm"})`

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